

FINAL EXAM**Fall 2023**

- This is a closed-book exam. Only a 8.5"x11" formula sheet is allowed.
- No electronic devices are allowed, except for the simple calculator we provide.
- You may not communicate with other students during the exam. You will also keep the exam contents confidential.
- There are **180 points in the exam**. You also have **180 minutes**.
- Do not be concerned if your answers are not whole numbers. Large numbers are avoided to lower the burden of calculation. Don't worry if you find the numbers unrealistic---you could make sense of these numbers in your head by picking the right units yourself (but no need to answer with these units).
- Please be as complete as possible. Write down the reasonings for your answer (unless instructed otherwise).
- Complete all **SHORT ANSWER** questions in one Blue Book, and all **LONG PROBLEMS** in a 2nd Blue Book. **Write your name and "SHORT" / "LONG" on the cover of your Blue Books.**
- Answers written **anywhere else, including on the question sheet** will **NOT** be graded.
- Please return the **Blue Books** AND **the question sheet** after the exam.
- A camera will be set in the classroom to monitor exam situations (Exam recordings will be deleted once final grades are posted unless required for an ongoing Honor Committee case).
- If you have clarifying questions but there is no proctor in your classroom, please find TAs at each of the "four wings (#2210,#2410,#4200,#4400)" of the building. Faculty will also circulate in each classroom throughout the exam.

Section I: SHORT QUESTIONS (72 points)

Please be concise. None of the answers require more than a brief paragraph. All questions are worth 9 points each.

- a) (9 points) The state of Illinois increased its tax rate on cigarette by almost 200% in 2019. Among smokers (assuming they don't quit immediately), non-smokers, and tobacco companies, which group would bear the highest percentage of tax incidents? Why?
- b) (9 points) Amazon's warehouse uses both machines and labor to produce packages. Suppose that the price of labor increases due to changes in the minimum wage regulations. Given that Amazon had the optimal (cost-minimizing) combination of labor and machines before the price change, how will Amazon respond following the price change? Why?
- c) (9 points) Charley's place in Evans Hall used to charge \$8 per meal pre-pandemic. It now charges \$12 per meal. They have found a 10% decrease in sales. What is the demand elasticity at the price of \$8 (approximately)? Was it pricing optimally before?
- d) (9 points) A friend claims that, while it is true that an import tariff is likely to increase prices to consumers, the country is not any worse off because the government collects the tariff revenue. Do you agree or not, and why?
- e) (9 points) Suppose that, after having done some market research, Tesla has identified two segments of car buyers with different willingness to pay: Luxury buyers, who have high reservation prices for a Tesla, and Eco buyers, who have lower reservation prices. Tesla wishes to price discriminate among them but cannot distinguish between Luxury and Eco buyers, so it is considering setting up a self-selecting pricing scheme to extract as much surplus as possible from its consumers. With such a pricing scheme, will Tesla be able to extract all possible surplus from its Luxury customers? What about from its Eco customers? Explain.
- f) (9 points) You are in the process of opening a new golf course in New Haven and are in charge of setting up prices. The marginal cost of each additional entry to the golf course is greater than zero. You are considering charging your customers a monthly fee, which allows unlimited entry to the golf course. Is the best pricing scheme you can adopt in this case? If yes, explain why. If not, please provide an alternative.
- g) (9 points) Does the introduction of a mandate in health insurance markets make everyone better off? Does it make the market efficient? Please briefly argument your answer.
- h) A local store selling handmade sweaters is currently paying its salespeople a fixed hourly wage, but sales are dwindling. The owners are brainstorming strategies to increase sales. In particular, they are discussing changing salespeople's compensation to a package that includes an hourly wage and a sale-contingent bonus. Is it worth it for the store to adopt this compensation scheme? What does your answer depend on?

Section II: LONG PROBLEMS (108 points)**Problem 1: Electric Vehicles (18 points)**

Electric Vehicles (EVs) have lower carbon footprints than gasoline-powered cars and do not produce as much fine particle matter pollution as combustion from gasoline/diesel car engines do. However, these environmental benefits are not directly taken into account when purchasing EV's. In other words, EVs have *positive externality*.

- a) (2 points) Without any market interventions, will EV consumption be lower or higher relative to the efficient level?

Suppose that, in the city of New Haven, the demand of gasoline-powered cars is given by

$$Q_D(p) = 40 - p,$$

Meanwhile, suppose that the supply of gasoline-power cars is given by

$$Q_S(p) = 4p$$

- b) (4 points) What is the equilibrium price and quantity of gas-powered cars in New Haven (without government intervention)? What are the demand and supply elasticities at this equilibrium? Which side is more elastic?

The city of New Haven decided to intervene the market and encourage consumers to switch from gas-powered cars to EVs. Suppose that the city imposes a per-unit tax on gas-powered cars in the amount of \$10.

- c) (4 points) What is the equilibrium price and quantity of gas-powered cars in New Haven after tax? What percentage of tax burdens do consumers bear?
- d) (3 points) What is the amount of deadweight loss in the gas-powered car market?
- e) (3 points) Suppose that after tax, all the gas-powered car consumers with reservation prices lower than the new equilibrium price switch to EVs. How many more EVs are purchased after tax?
- f) (2 points) Overall, what can you say about this policy?

Problem 2: Gravel market (25 points)

Quarry Inc. has a local monopoly over the gravel industry in central Connecticut. The inverse demand for gravel is given by $Q = 100 - 2P$, where Q is the quantity of gravel in thousand cubic yards per month, and P is the price in dollars per cubic yard. Quarry Inc. incurs a fixed cost of production of \$12 and a variable cost of \$5 per cubic yard.

- a) (4 points) What are the profit maximizing price and quantity choices for Quarry Inc.?
- b) (4 points) What is the elasticity of demand at this price? Does this elasticity make sense to you? Why or why not?
- c) (2 points) What is the total profit of Quarry Inc. at the optimal price and quantity choice?
- d) (3 points) If the market were instead perfectly competitive, with many gravel providers (all with identical marginal costs as the monopolist), what would you predict the price of gravel to be? What would be the quantity sold?
- e) (3 points) What would the firm's profit be in this scenario? What would the firm do in the long vs short run?
- f) (4 points) Let's go back to the setting with Quarry Inc. being a monopoly (i.e., part (d) no longer applies.) The CT state government believes that local builders are suffering because of Quarry Inc.'s monopoly power and consequent high prices. To extract some surplus from the monopoly, the government levies a \$1 tax (to be paid by Quarry Inc.) for every cubic yard of gravel sold. Compute the new price and quantity that you expect in the market.
- g) (5 points) Do you think the above tax is effective in helping the local builders? Explain quantitatively why or why not.

Problem 3: Job offers (20 points)

Knowing you have taken Basics of Economics, a non-MBA friend who is deciding between job options has come to you for advice. They have been offered two jobs, one that has a guaranteed salary of \$120,000 per year, and the other that has a lower salary of \$100,000 per year but has a 75% chance of also having a \$60,000 bonus.

This student is risk-averse and so is considering both opportunities. Their utility function is represented by $u(x) = \sqrt{x}$ where x is measured in thousands so that $\sqrt{\$100k} = 10$. They have no other savings or wealth.

- a) (2 points) What is the expected value of the two jobs?

- b) (4 points) Which job do you expect them to take?
- c) (2 points) Would your answer change if your friend told you they are risk-neutral? Why or why not?
- d) (6 points) What is the minimum guaranteed salary the first job would have to offer to make the student indifferent between the two jobs?
- e) (6 points) The firm offering the bonus has decided to offer a different contract for new analysts to accept. This new contract maintains the Expected Value of the existing contract but puts more weight on the bonus. Is this a good idea or bad idea when hiring potentially risk-averse employees, and why?

Problem 4: An extended HFT investment game (25 points)

Consider an extended version of the high-frequency-trading game discussed in the class. Suppose now that when a firm invests, it can choose whether to do a high investment (HI) or a low investment (LI). The matrix form of this extended game is as follows, where “No” means no investment and X and Y are unknown numbers:

		Firm 2		
		HI	LI	No
Firm 1	HI	30, 30	X, 25	60, 20
	LI	25, X	40, 40	Y, 35
	No	20, 60	35, Y	50, 50

- a) (8 points) For what values of X and Y, will HI be a dominant strategy for each firm?
- b) (8 points) For what values of X and Y, will this game have more than one Nash equilibrium?
- c) (9 points) For what values of X and Y, will firm 1 earn 35 in the **sequential-move** version of this game where firm 1 moves first?

Problem 5: An extended location choice game (20 points)

Consider an extended version of the location choice game discussed in the class. Tourists are evenly distributed on a beach that is a line segment $[0,1]$ of one mile long. **THREE** ice cream vendors simultaneously choose where to locate, each aiming to maximize their own market share. (Suppose they can locate exactly at the same spot if they want.) Tourists prefer to buy from the closer vendor. When they are indifferent between multiple vendors, they randomly choose one.

- a) (5 points) Is it a Nash equilibrium that the three vendors all locate at the middle of the beach (i.e., at $1/2$)? Please explain your answer.
- b) (5 points) Is it a Nash equilibrium that the three vendors locate at $1/6$, $1/2$, and $5/6$, respectively (so that each has a market share $1/3$)? Please explain your answer.
- c) (5 points) Suppose now we “bend” the beach so that it becomes a **CIRCLE** that has a circumference of one mile (i.e., the two endpoints 0 and 1 are now connected to each other). Is the location configuration in question (b) a Nash equilibrium? Please explain your answer.
- d) (5 points) Return to the original case where the beach is a line segment. Suppose that vendor 1 has already entered the market and located at the left endpoint 0. Now vendors 2 and 3 simultaneously choose where to locate. What is the Nash equilibrium of the location choice game between vendors 2 and 3?